📘 Medical Imaging Evaluation Handbook

(For X-ray Fracture Detection & Beyond)

🧠 0. First Principle (Don't Skip This)

Every metric answers **one of three questions**:

1. **Discrimination** → Can the model separate classes?
2. **Calibration** → Are predicted probabilities trustworthy?
3. **Localization** → Is the model looking at the correct region?

If your evaluation doesn't cover all three → incomplete.

🧩 1. Problem Types (Metrics depend on this)

Task	Example	Metrics
Classification	Fracture vs Normal	Accuracy, Recall, AUC
Detection	Find fracture location (box)	mAP, IoU
Segmentation	Pixel-wise fracture region	Dice, IoU
Multi-label	Multiple conditions	Macro/micro metrics

🧮 2. Confusion Matrix (Core of Everything)

	Predicted Positive	Predicted Negative
Actual Positive	TP	FN
Actual Negative	FP	TN

🧠 Interpretation

- TP → Correct detection
- FN → Missed disease (**critical**)
- FP → False alarm
- TN → Correct rejection

👉 In medicine: **FN is usually the worst**

📊 3. Classification Metrics (Complete Set)

✅ Accuracy

[
\frac{TP + TN}{Total}
]

- Misleading in imbalanced datasets

🎯 Precision (Positive Predictive Value)

[
\frac{TP}{TP + FP}
]

👉 “When model says fracture, how often correct?”

🔍 Recall (Sensitivity)

[
\frac{TP}{TP + FN}
]

👉 “How many real fractures did we detect?”

🔥 Most critical in diagnosis

🛡️ Specificity

[
\frac{TN}{TN + FP}
]

👉 “How well we avoid false alarms”

⚖️ F1 Score

[
\frac{2PR}{P + R}
]

👉 Balance of precision & recall

📌 Balanced Accuracy

[
\frac{Sensitivity + Specificity}{2}
]

👉 Useful for imbalanced data

📊 MCC (Matthews Correlation Coefficient)

[
MCC = \frac{TP×TN - FP×FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}
]

👉 Best single metric for imbalance

- +1 = perfect
- 0 = random

🎯 Cohen’s Kappa

Measures agreement beyond chance

👉 Useful when comparing with radiologists

📈 4. Threshold-Based vs Threshold-Free Metrics

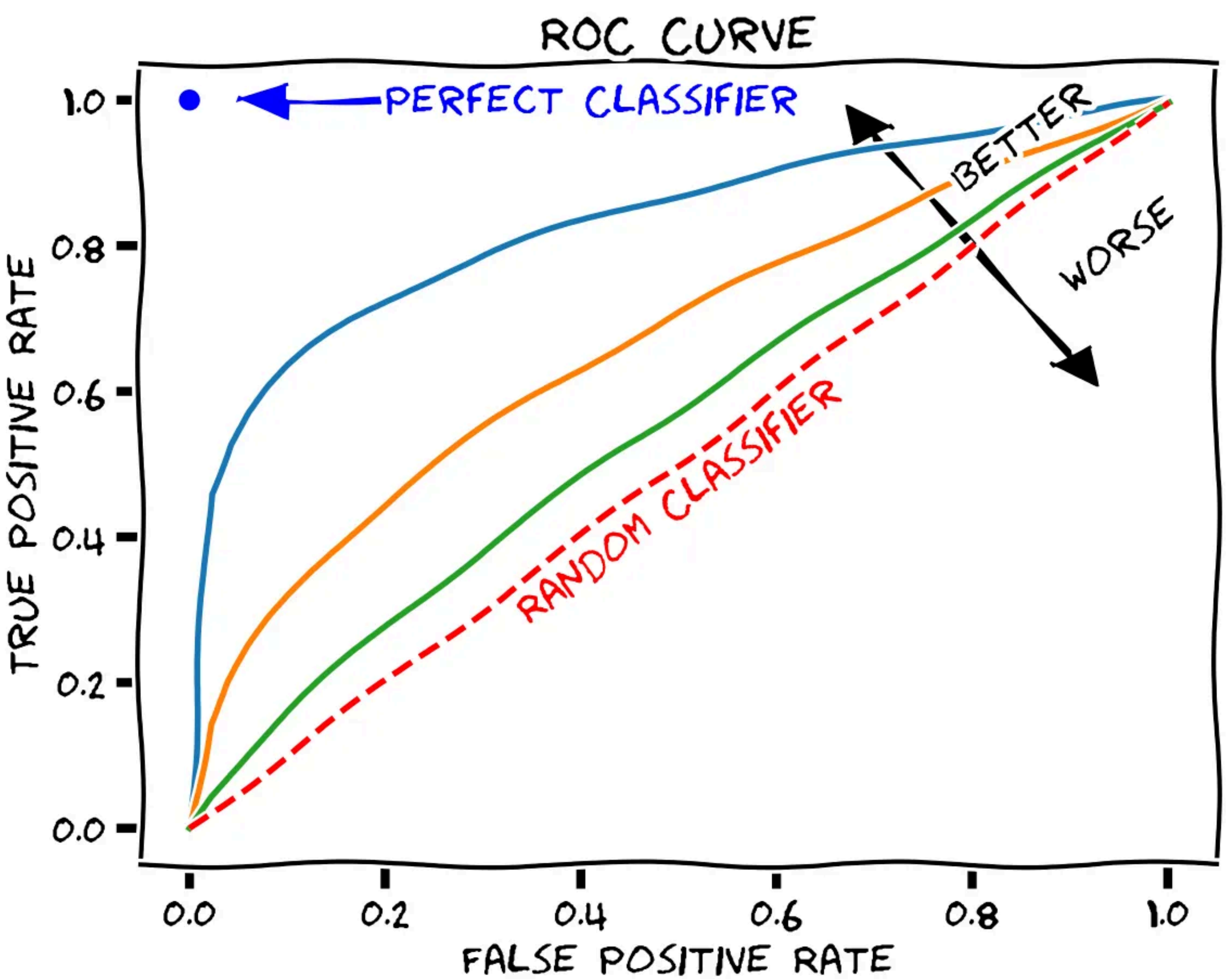
🔥 ROC Curve & AUC

- Measures **ranking ability**

Interpretation:

- AUC = probability model ranks positive higher than negative

Image




```
error code: 1020
```

📉 Precision-Recall Curve

Better when:

- Dataset is imbalanced (which yours is)

⚠️ Key Difference

Metric	Use Case
ROC-AUC	Balanced datasets
PR-AUC	Imbalanced datasets

📏 5. Calibration Metrics (Almost Everyone Ignores This)

This is where most “high accuracy models” fail clinically.

🧠 What is Calibration?

If model says:

- “80% fracture”

👉 Is it actually correct 80% of the time?

📊 Metrics

🔹 Brier Score

[
Mean (predicted - actual)^2
]

- Lower is better

🔹 Expected Calibration Error (ECE)

Measures mismatch between:

- predicted probability vs actual outcome

🔹 Reliability Diagram

Image

Image

```
error code: 1020
```

🧠 Interpretation

Pattern	Meaning
Below diagonal	Overconfident
Above diagonal	Underconfident

🧩 6. Detection Metrics (Bounding Boxes)

Used if you localize fractures.

📏 IoU (Intersection over Union)

[
\frac{Overlap}{Union}
]

🎯 mAP (Mean Average Precision)

- Measures detection performance across thresholds

🧠 Interpretation

- mAP@0.5 → loose localization
- mAP@0.75 → strict

📐 7. Segmentation Metrics (Pixel-Level)

🎯 Dice Coefficient

[
\frac{2TP}{2TP + FP + FN}
]

👉 Most used in medical imaging

📏 IoU (Jaccard Index)

[
\frac{TP}{TP + FP + FN}
]

🧠 Difference

- Dice is more forgiving
- IoU is stricter

🩻 Segmentation Example

Image

```
error code: 1020
```

Image

🔍 8. Localization Metrics (Weak Supervision / Grad-CAM)

📌 Pointing Game

- Check if peak activation lies inside fracture region

📏 Localization Accuracy

- % of heatmaps overlapping annotated region

🎯 Energy-Based Metrics

- How much heatmap mass lies inside ground truth

🧠 9. Explainability Metrics (Grad-CAM & Beyond)

📌 What to Evaluate

1. Faithfulness

- Does heatmap reflect model decision?

2. Localization

- Does it align with pathology?

3. Robustness

- Does explanation change drastically with small noise?

⚠️ Reality

- Grad-CAM is **not reliable alone**
- Compare with:
  - Grad-CAM++
  - Score-CAM
  - Eigen-CAM

🧪 10. Statistical Evaluation (You NEED this for PhD level)

📊 Confidence Intervals

- Report uncertainty in metrics

🧠 Bootstrapping

- Resample dataset → compute metric distribution

🧪 Hypothesis Testing

- Compare models (e.g., AUC difference)

📉 11. Error Analysis Metrics

📌 Per-Class Metrics

- Fracture types (wrist, hip, etc.)

📌 Hard Case Analysis

- Small fractures
- Low contrast
- Occlusions

📌 Subgroup Analysis

- Age groups
- Imaging views

📊 12. Training Metrics (Model Behavior)

📉 Loss Curve

- Optimization progress

📈 Accuracy Curve

- Learning behavior

🚨 Red Flags

Pattern	Problem
Train ↑ Val ↓	Overfitting
Both low	Underfitting

🧠 13. Clinical Metrics (Rarely Used but Powerful)

📌 NPV (Negative Predictive Value)

[
\frac{TN}{TN + FN}
]

👉 "If model says normal, can we trust it?"

📌 PPV (Same as Precision)

📌 Decision Curve Analysis

- Measures clinical usefulness

📦 14. Generalization Metrics

📌 Cross-Dataset Performance

Train → MURA
Test → FracAtlas

👉 Measures robustness

📌 Domain Shift Sensitivity

- Performance drop across datasets

🧠 15. What a Complete Evaluation Looks Like

✔ MUST HAVE

- Confusion matrix metrics
- ROC + PR curves
- Calibration analysis
- Error breakdown

✓ STRONG PAPER

- Localization metrics
- Explainability validation
- Cross-dataset testing
- Statistical significance

✓ TOP-TIER RESEARCH

- Clinical validation
- Human vs AI comparison
- Failure mode taxonomy

🚫 Final Reality Check

If your evaluation:

- Ignores calibration → not clinically usable
- Ignores FN → dangerous
- Ignores localization → not trustworthy
- Ignores generalization → not publishable

🧠 Bottom Line

- **Metrics ≠ understanding**
- Each metric answers a different failure mode
- Real research = connecting them into a **coherent story**